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Synopsis on

"Smart Glove for Sign Language Translation and Gesture Controlled Home Automation"

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"Smart Glove for sign language translation and Gesture controlled home automation"

1. ABSTRACT

The Smart Glove system is a wearable assistive device designed to translate common sign-language gestures into audio and text messages while simultaneously enabling gesture-based control of domestic appliances. The glove integrates an **ADXL345 3-axis accelerometer** mounted on a wearable glove, an **Arduino** microcontroller for signal processing, an **I2C LCD** for visual output, and a **DF Player Mini** MP3 module for voice playback. Predefined ranges of acceleration (primarily X-axis gesture signatures) are used to classify gestures mapped to phrases like "Emergency," "I need help," and "I am hungry," and to appliance-control commands such as "Light ON/OFF" and "Fan ON/OFF." Relays drive appliance loads. The system offers an intuitive, low-cost, and portable solution to support people with hearing or speech impairments: translating gestures into understandable audio/text and enabling independent control of basic home appliances. The project demonstrates hardware interfacing, real-time sensor processing, embedded decision logic, and basic user feedback mechanisms. Future work includes expanding gesture vocabulary, adding machine-learning-based classification for robustness, wireless connectivity (Bluetooth/Wi-Fi), and smartphone integration for logging and remote monitoring.

2. KEYWORDS

Smart Glove, Sign Language Translation, ADXL345, Gesture Recognition, DF Player Mini, Arduino, Home Automation, Relay, I2C LCD, Assistive Technology

3. INTRODUCTION

Communication and independent living are major challenges for people with hearing and speech impairments. Wearable gesture-recognition devices — smart gloves — can translate hand/arm gestures into audible and textual output and can be used to control home appliances without voice or touch. This project develops a prototype smart glove that recognizes a set of predefined gestures using inertial sensing (ADXL345 accelerometer) and maps each gesture to either an audio/text phrase or a home automation command. The goal is an affordable, portable assistive system usable in everyday home environments.

Objectives:

Primary objectives

- Build a wearable smart glove to capture hand gestures using ADXL345 accelerometer.
- Translate predefined gesture patterns into textual messages (I2C LCD) and audio (DFPlayer Mini).
- Implement gesture-controlled home automation toggling (light and fan) using relay modules.

Secondary objectives

- Ensure real-time operation and low latency gesture recognition.
- Make the system portable, user-friendly and power-efficient.
- Provide clear documentation, test cases, and demonstration scenarios.

4. LITERATURE SURVEY

- Wearable gloves using flex sensors and IMUs to detect finger motions and sign language translation.
- Systems combining accelerometers (ADXL345/MPU6050) and pattern matching for gesture classification.
- Assistive translation systems that use recorded audio outputs or TTS modules for communication.
- Gesture-controlled home automation using hand orientation and motion sensors with relays/Bluetooth relays.

5. EXISTING SYSTEM

- Use only flex sensors (finger bends) — limited to finger-based alphabets but less robust for whole-hand gestures.
- Require complex ML and large datasets (increasing cost and processing).
- Lack integrated home automation control.
- Are not easily portable or have short battery life.

6. PROPOSED SYSTEM

- ADXL345 accelerometer for robust motion/tilt detection (mounted on glove back).
- An Arduino (Uno/Nano) to sample acceleration, apply threshold-based gesture classification, and trigger outputs.
- I2C 16×2 LCD for visual text output.
- DF Robot DF Player Mini for playback of pre-recorded phrases.
- Relay modules to switch household appliances (light, fan).
- Software: simple threshold/range decision logic (as in prototype code) for reliable and low-latency classification.

Advantages:

- Low computation requirement (no heavy ML), hence simple and robust.
- Portable, battery-powered, and inexpensive.
- Easy to calibrate per-user by adjusting thresholds.

7. REQURIMENTS SPECIFICATION

Functional Requirements

- Real-time sampling of ADXL345 at suitable rate (e.g., 50–100 Hz).
- Gesture-to-action mapping table (10 gestures):
    1. Fan OFF
    2. Fan ON
    3. Light ON
    4. Light OFF
    5. Emergency
    6. I am Hungry
    7. What
    8. I need Help
    9. Hello
    10. I don’t understand
- Clear LCD message display for every recognized gesture.

- Audio playback of corresponding pre-recorded MP3 files.
- Relay actuation safety (isolation and appropriate drivers).

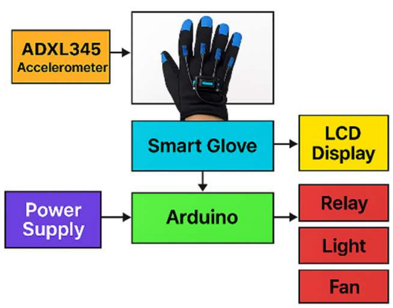
Non-functional Requirements

- Latency: recognition and response within 0.5–2 seconds.
- Usability: glove should be comfortable for short durations.
- Portability: system must be battery powered (5–9V) and lightweight.
- Safety: relays must be chosen to match load; mains wiring must follow safety rules.

8. METHADOLOGY:

- Initialize peripherals (ADXL345, LCD, DF Player, relay pins).
- Continuously read accelerometer data (X, Y, Z).
- Optionally apply basic filtering (moving average) to reduce noise.
- Compare filtered X-axis (and/or Y/Z if needed) to predefined ranges; map to gesture IDs.
- For each recognized gesture:
- Update LCD with textual message.
- Play corresponding MP3 via DF Player.
- If gesture maps to appliance control, toggle corresponding relay pin.
- Add debouncing / cooldown to prevent repeated triggers (e.g., 2–3 s lockout).
- Provide a calibration routine at start to set neutral/rest thresholds per user.

9. SYSTEM ARCHITECTURE:



Smart Glove (hardware):

- ADXL345 accelerometer (I2C) mounted on glove
- Arduino Uno / Nano (microcontroller)
- I2C 16×2 LCD
- DFPlayer Mini + speaker
- Relay module (2 channels)
- Power supply (Li-ion or 5V regulator)

Software blocks:

- Sensor interface (I2C read)
- Preprocessing & filtering (simple moving average)
- Gesture decision logic (threshold ranges on acceleration)
- Output manager (LCD text, DFPlayer audio, relay control)

7. EXPECTED RESULTS

- Working prototype smart glove with at least 8–10 gesture mappings.
- Demonstration videos and documentation.
- Project report (Phase-1/Phase-2 as required).
- Source code and hardware wiring diagram.
- User manual & calibration instructions.

8. CONCLUSION

The Smart Glove system effectively translates basic hand gestures into audio and text outputs while also enabling gesture-based control of home appliances. This low-cost and portable solution supports differently-abled individuals by improving communication and promoting independent living. With further enhancements, the system can evolve into a more advanced and fully scalable assistive technology.

9. REFERENCES

- **R. Kadam, P. Pansare, P. Gawali, and S. Bhosale**, “Smart Glove for Sign Language Recognition,” *International Journal of Advanced Research in Computer and Communication Engineering*, vol. 6, no. 4, pp. 388–392, 2017.
- **A. S. Jadhav, V. H. Patil, and S. R. Khot**, “Hand Gesture Recognition Based on Flex Sensors for Sign Language Translation,” *International Journal of Engineering Research & Technology (IJERT)*, vol. 9, no. 6, 2020.
- **J. Liu, Y. Jiang, and L. Zhang**, “Real-Time Continuous Gesture Recognition for Natural Human-Computer Interaction,” *IEEE Transactions on Multimedia*, vol. 16, no. 8, pp. 2281–2291, Dec. 2014.
- **S. R. Kamat and A. S. Shinde**, *Embedded Systems and Robotics with Open-Source Tools*, CRC Press, 2021 – (for ESP32/Arduino integration and embedded systems design).
- **DFRobot**, *DFPlayer Mini MP3 Module Wiki and Documentation*, Available online: https://wiki.dfrobot.com/DFPlayer_Mini_SKU_DFR0299.